

Housing Processor — Master Overview

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Audience: anyone who needs to understand what this system is, what it does today, and where it's going — without reading the code.

In one sentence

This is an **internal tool for housing staff** that turns student housing Word applications into organized roommate-group records (and, later, Excel reports) — with a database as the single source of truth.

Why it exists

Today, student housing applications often arrive as Microsoft Word (.docx) files at different times. Roommate names are messy, preferences vary, and staff spend a lot of time copying information into spreadsheets.

This system is meant to:

1. **Receive and keep** the original Word files safely
2. **Extract** applicant, roommate, contact, and property preference details
3. **Group** people into roommate groups with stable group numbers
4. **Flag** anything unclear for a human to decide
5. **Export** a clean Excel workbook for day-to-day operations

Students do **not** use this product. It is for **operations / housing staff** only.

The big idea (non-negotiable)

Thing	Role
Database	The official record — what the business trusts

Word documents	Inputs only — kept as archives, not edited as the master list
Excel workbooks	Reports generated <i>from</i> the database — never the master list

Automation may help read a document or suggest a match. It should **not** silently invent group numbers or overwrite conflicting data without staff review when something is ambiguous.

Where the app stands today

Think of the current product as a **working scaffold**: the doors, rooms, and wiring are in place, but several rooms are still empty.

What staff can do now

- **Sign up / sign in** with email and password (when connected to Supabase)
- **Browse** the ops screens without being forced to a login page first
- **Upload** a housing application Word file from the Dashboard
- **See uploaded applications** on the Applications page
- **Browse Groups** (list and detail) — these stay empty until real grouping is built
- Have the system **start processing** an application (via a staff/developer action); it currently ends in “**needs review**” because automatic extraction and matching are not finished yet

What is not ready for daily use yet

Capability	Status in plain English
Reading fields out of Word (name, email, roommates, houses, GPA, etc.)	Placeholder — text can be read, but structured mapping is not done
Automatically putting people into roommate groups	Not built — unclear cases always go to “needs review”

Creating or editing groups in the UI	Not built
A real review queue (decide match / fix data)	Not built
Generating and downloading Excel for ops	Not wired end-to-end
Email inbox ingest, CRM sync, student portal	Future work

Bottom line: staff can already get files into the system and see them listed. They **cannot yet** rely on the system to extract fields, assign group numbers, resolve reviews, or produce the Excel workbook. Closing that gap is the path from “scaffold” to “daily tool.”

What you see on screen

The staff website and the behind-the-scenes service run together (typically at `http://127.0.0.1:8000/` when running locally).

Screen	What it's for
Dashboard	Check that the system is up; upload a .docx application
Groups	Browse roommate groups (filter by group number)
Group detail	See members of one group (contact person, email, phone)
Applications	See every uploaded file: name, status, when it arrived
Sign in / Sign up	Create an account or log in with email and password

The top bar always offers **Dashboard**, **Groups**, **Applications**, a **group number** search, and either **Log in / Sign up** or your email plus **Log out**.

The staff journey (today vs intended)

Today's realistic journey

1. Open the site 2. Sign in (or sign up) if you need to upload or load lists 3. Upload a Word application on the Dashboard 4. Confirm it appears under Applications 5. Processing (when run) marks it "needs review" 6. Groups stay empty until grouping features are built

The intended full journey (once Phase 1 is complete)

Word application arrives ↓ Staff uploads it (or email ingest, later) ↓ System stores the file and creates an application record ↓ System extracts applicant + roommate + preference details ↓ System tries to match to an existing roommate group ↓ Clear match → attach to that group Ambiguous / new → staff review → create or attach group ↓ Group numbers are permanent and sequential ↓ Staff exports Excel from the database whenever needed

How information moves (plain English)

1. **Someone uploads a Word file.**
2. The system checks whether that exact file was already uploaded (duplicate detection).
3. The original file is **archived**; a row is written in the **database** saying an application was received.
4. When processing runs, the system tries to understand the document and decide on a group.
5. Today that decision almost always becomes "**a human needs to look at this.**"
6. Later, staff will resolve those cases, groups will fill in, and Excel will be generated from the database — not by retyping Word into a spreadsheet.

Accounts and access

- Staff create an account with **email and password** (Sign up) or use an existing one (Log in).
- After login, the site **remembers you** as you move between pages.

- Looking around without logging in is allowed; **loading lists or uploading** requires being signed in.
- Only **active staff accounts** can use the protected features.
- This is **not** a public student portal.

What's underneath (only as much as you need)

You do not need this to use the product, but it helps when talking to developers:

- A **staff website** (simple web pages)
- A **service** that accepts uploads, talks to the database, and enforces login
- A **database** (cloud Postgres via Supabase, or a local file for offline experiments)
- **Login** handled by Supabase Auth
- Libraries that can read Word and (later) write Excel

The important product rule stays the same regardless of tools: **database first; Word in; Excel out.**

Roadmap in plain language

Work is organized so the system becomes useful for daily ops **before** fancy matching or AI.

1. **Reliable reading of Word applications** into structured fields
2. **Saving applicants** and creating real “needs review” items when something is unclear
3. **Manual group tools** — create a group or attach someone to an existing one (with permanent group numbers)
4. **Excel export** that staff can download from live database data
5. **Smarter matching** (and optional AI assist) only after the above is trustworthy

Later phases may add email intake, CRM connections, richer review screens, and operational hardening (backups, metrics, etc.).

How to talk about status with others

Phrase	Meaning
“We can ingest applications”	Upload + store + list works
“We’re not live for matching yet”	Groups and auto-assignment are not production-ready
“Review is a placeholder”	Processing stops at “needs review”; there is no full review workspace
“Excel is next after extraction and groups”	Export depends on having real data in the database
“Scaffold, not finished ops tool”	Foundation is real; Phase 1 product outcomes are still open

Related documents

Document	Who it’s for
This file (MASTER_OVERVIEW.md)	Everyone — shared understanding
README.md	Developers — how to install and run
cursor-docs/ (local only)	Detailed design notes for builders; not required for business readers

Closing

Housing Processor is being built so housing staff can stop treating Word and Excel as the system of record.

Today: sign-in, upload, and application listing are real; processing stops at human review; grouping and Excel are still ahead.

Next: make extraction and group workflows trustworthy enough that Excel can be regenerated from the database on demand — then the tool becomes something operations can rely on every day.